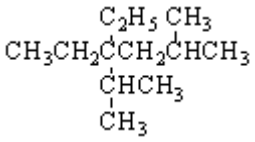
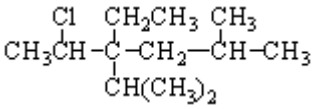
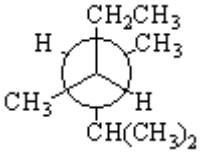
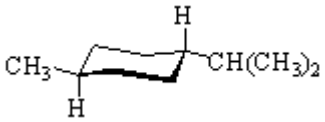
单元自测题 Ⅰ

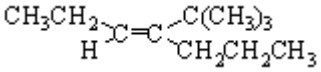
1. 用系统命名法命名下列化合物:

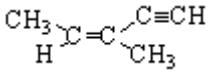
(1). 

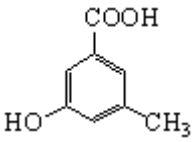
(2). 

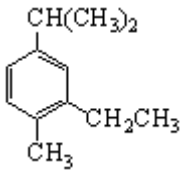
(3). 

(4). 

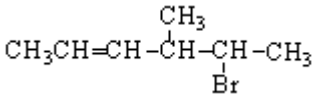
(5). 



(7). 

(8). 

(9). $(CH_3)_3CC(CH_3)_3$

(10). 

2. 写出下列各化合物结构式:

(1). Z-3,4-二甲基-2-己烯

(2). E-3-甲基-4-异丙基-3-辛烯

(3). 2,2,5-三甲基-4-乙基己烷

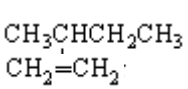
(4). 2-甲基-4-氨基苯磺酸

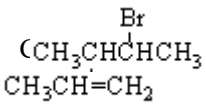
(5). 3,4-二甲基-1-环戊基-5-氯己烷

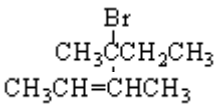
(6). 4-甲基-2-戊炔

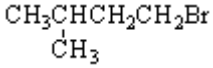
3. 选择填空:

(1). 在光照条件下异戊烷与溴反应时, 生成下列一溴代烷, 其中含量最高的是( )

A. 



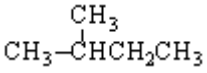


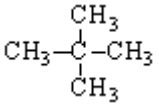


(2). 下列化合物与HBr发生加成反应时, 速度最快的是(C)

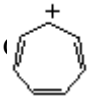
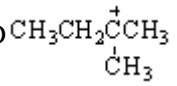
A.  B. C. D.

(3). 下列化合物中沸点最高的是(　), 熔点最高的是(　)

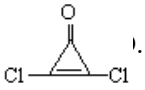
A. $CH_3CH_2CH_2CH_2CH_3$　　 

C.  　D.　　$CH_3CH_2CH_2CH_3$

(4). 下列碳正离子中间体最稳定的是(　)

A. B.  　D  　 

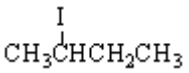
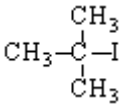
(5). 下列结构中没有芳香性的是(　)

A.  B.  　  　 

(6). 卤代烃水解反应历程属于(　)

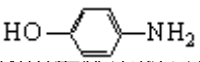
A. 亲电加成　B. 亲电取代　C. 亲核取代　D. 游离基取代

(7). 下列化合物按S_N2取代反应速度最快的是(　)

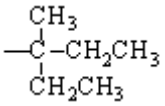
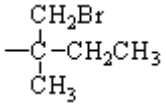
A. $CH_3CH_2CH_2CH_2Br$　C $CH_3CH_2CH_2CH_2I$　  　 

(8). 下列结构中同时存在π－π、p-π、σ－π共轭的是(　)

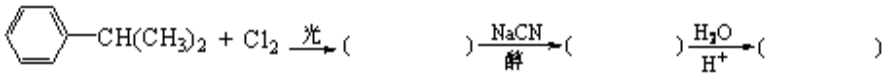
A. $CH_3CH=CH-CH_2Cl$　　$Cl-CH=CH-CH_2-CH=CH-CH_3$

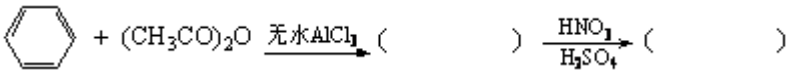
C.  　D.　 

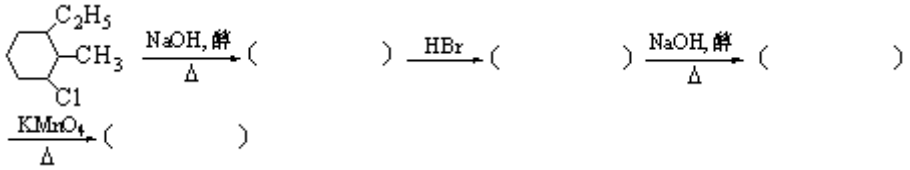
(9). 根据贝采尔-凯恩优先次序规则, 下列基团中最优先的是(　)

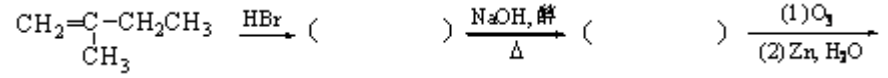
A. $-CH_2OH$　C. $-C(CH_3)_3$ D.  　 

4. 完成下列反应式:

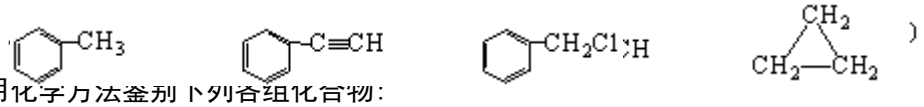
(1). 

(2). 

(3). 

(4). 

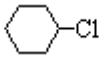
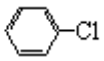
(　　) + (　　)

(5). 

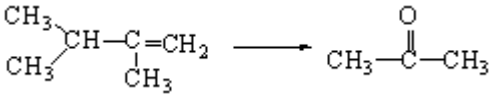
5. 用化学方法鉴别下列各组化合物:

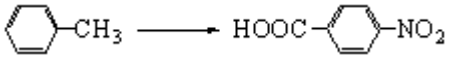
(1).

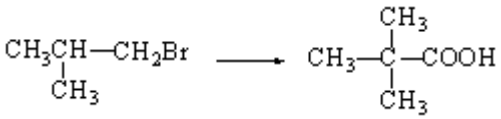
(3). $CH_3CH_2CH_2CH_2I$ $CH_3CH_2CH_2CH_2Cl$ $CH_3CH_2CH_2CH_2Br$

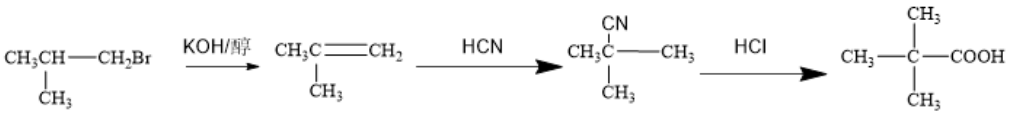
(4). $CH_3CH{=}CHCH_2Cl$   $CH_2{=}CHCl$

6. 完成下列转化（无机试剂任选）:

(1). 

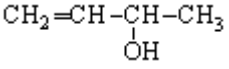
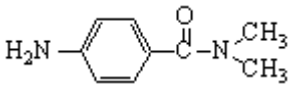
(2). 

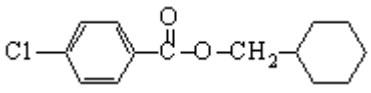
(3). 

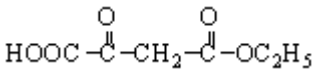


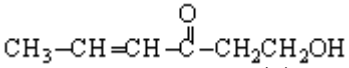
单元自测题 II

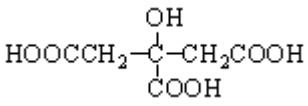
1. 命名下列化合物或写出结构式:

(1).  (2). 

(3). 



(5). 

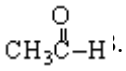
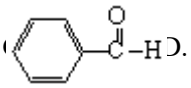
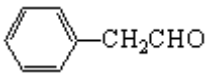


(7). 苹果酸酐　　(8). 氯化三甲基苯基铵

(9). 2-甲氧基对苯二酚　　(10). 2,6-二甲基−1,4-萘醌

2. 选择题:

(1). 下列化合物中能发生羟醛缩合反应的是( ), 能发生歧化反应的是( ):

A.  　B.  　C.  　D. HOOC−CHO

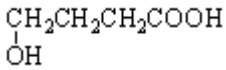
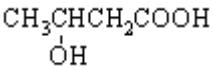
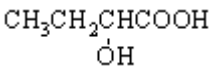
(2). 具有相同分子式的外消旋体与内消旋体之间是( )。

A. 对映体　B. 非对映体　C. 差向异构体　D. 互变异构体

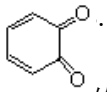
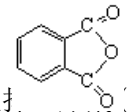
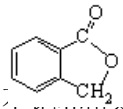
(3). 下列干燥剂中不能用于干燥低级醇的是( ):

A. 氧化钙　B. 无水氯化钙　C. 无水硫酸钠　D. 无水硫酸镁

(4). 下列三种羟基酸, 加热脱水产物为不饱和酸的是( ), 产物为内酯的是( ), 产物为交酯的是( ):

A.  　C  　 

(5). 下列化合物中( )属于醌类化合物, ( )属于环二酮, ( )属于酸酐, ( )属于内酯:

A.  　  　D.  　 

(6). 欲将不饱和酸还原为不饱和醇, 应选择还原剂( ), 欲将不饱和酸还原为不饱和醇, 应选择还原剂( ), 欲将不饱和酸还原为饱和醇, 应选择还原剂( ):

A. H_2 / Ni　B. $NaBH_4$ / H_3^+O　C. $LiAlH_4$ / H_3^+O

(7). 3-戊酮酸乙酯与4-戊酮酸乙酯可以用( )来鉴别:

A. 2,4-二硝基苯肼　B费林试剂　C. 土伦试剂　D. $FeCl_3$

3. 下列说法你认为对的打“√”, 错的打“×”:

(1). 丙二烯分子是平面结构。　( )

(2). 缩二脲反应可用于鉴别含有酰胺键的化合物　( )

(3). 乙醇和乙醚是官能团异构体　( )

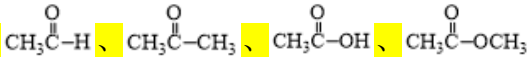
(4). 羧酸分子中烃基上的氢原子被其他官能团取代的化合物为取代酸, 而羧基中的羟基被其他原子或基团取代的化合物为羧酸衍生物　( )

(5). 羰基的加成反应为亲核加成反应　( )

(6). 可以用土伦试剂或费林试剂鉴别所有的醛和酮　( )

(7). 内消旋体和外消旋体都是混合物　( )

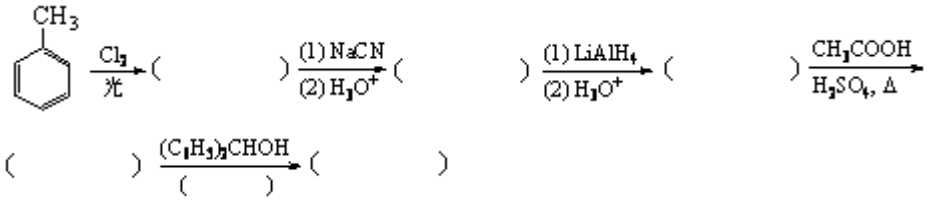
(8). 乙酰胺中含氨基所以显碱性　( )

(9).  都能发生碘仿反应　(×)

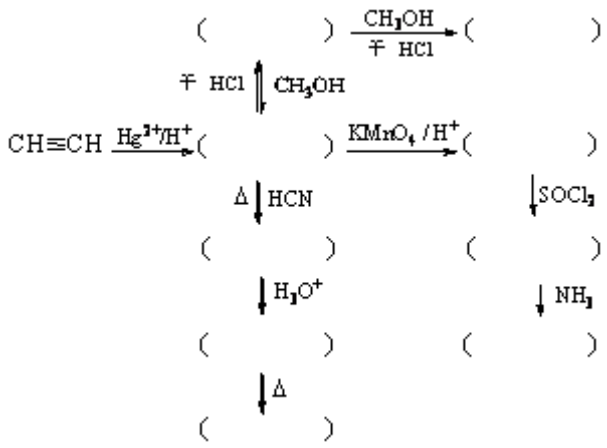
(10). 苯甲酰氯、邻苯二甲酸酐、苯甲酸乙酯、苯乙酰胺均能发生水解反应（ ）

4. 完成下列转化：

(1).

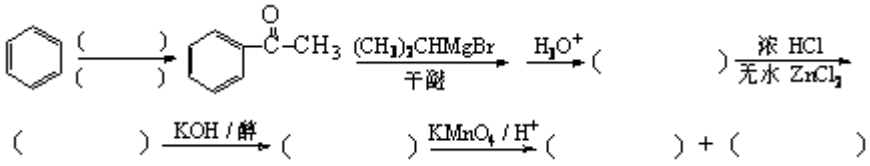


(2).

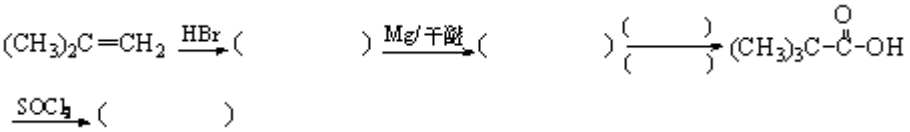


(3). $CH_3CH_2CHO \xrightarrow{\text{稀NaOH}} (\quad) \xrightarrow{\Delta} (\quad) \xrightarrow{HBr} (\quad)$

(4).

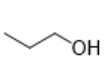


(5).



5. 鉴别、分离、和提纯：

(1). 用简单的化学方法鉴别下列化合物：

		CH_3CH_2CHO		CH_3CH_2COOH
$Ag(NH_3)_2^+$	(-)	$Ag\downarrow$	(-)	(-)
$I_2/NaOH$	(-)		$CHI_3\downarrow$	(-)
卢卡斯试剂	加热后混浊			(-)

丙醇 丙醛 丙酮 丙酸

(2). 用简单的方法分离下列各化合物：

环己醇　苯甲酸　环己酮　苯酚

(4). 甘油　淀粉　果糖　麦芽糖

淀粉	蓝色		
甘油	(—)	(—)	
果糖	(—)	紫色	红色
麦芽糖	(—)	紫色	(—)